

6(a)	any three from: nystose is not branched / glycogen is branched ; different glycosidic bonds, qualified ; e.g. nystose has one type of glycosidic bond <u>and</u> glycogen has two types 1-4 <u>and</u> 1-6 (glycosidic) bonds in glycogen (v 1-2 glycosidic bonds in nystose) two types of monomer / AW, in nystose v only glucose in glycogen ; nystose contains one glucose (residue / monomer) ; <i>ref. to ring shape</i>, glycogen monomers have six-sided rings <u>and</u> nystose has (a) six-sided ring and five-sided rings ; AVP ; e.g. fructose in nystose <u>and</u> no fructose in glycogen I numbers of C,H and O	3
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Question	Answer	Marks
6(b)	<i>any four from:</i> <i>must attempt both to gain max 4</i> <i>rough endoplasmic reticulum</i> translation (of mRNA) / polypeptide synthesis, at ribosomes ; A protein synthesis polypeptide folding (in lumen) ; A protein folding packaged in / formation of, transport vesicles / vesicles for transport, to Golgi body ; <i>Golgi body</i> packaging into, Golgi / secretory, vesicles ; A formation of, Golgi / secretory vesicle, containing glycoprotein A packaging into vesicles for transport to cell surface membrane <i>rough endoplasmic reticulum or Golgi body</i> (modification by) glycosylation (of protein) / described ; e.g attachment of, sugars / oligosaccharide / polysaccharide / carbohydrate <i>ref. to</i> checking for, misfolded / incorrectly synthesised, proteins ; additional detail of post-translational modification ; e.g. <i>suggestion of</i> forming quaternary structure addition of non-protein groups	4

Question	Answer	Marks
6(c)	<i>any one from:</i> cell signalling or receptor(s) / binding site(s) , for , chemicals / hormones / molecules ; cell recognition or (cell surface) antigens or cell-cell interactions ;	1